

**5. Division of Matrices**

Division of matrices is done by multiplying the numerator by the inverse matrix of the denominator.

Let: A and B are matrices

$$\frac{A}{B} = AB^{-1}$$

where:

B^{-1} is called the inverse matrix of matrix B

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Transpose Matrices

If matrix A is reflected in its main diagonal, so that all rows become columns and all columns become rows without changing their relative order of entries in the rows and columns, the result is a **transpose matrix, A^T** .

Hence, the transpose of matrix A with a size of $m \times n$ is denoted by A^T with a size of $n \times m$. The transposed matrix would contain elements where the first row is the first column of the original matrix, the second row is the second column of the original matrix and so on.

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Example:

$$A = \begin{bmatrix} -4 & 1 & 3 \\ -1 & 2 & 5 \end{bmatrix}$$

$$A^T = \begin{bmatrix} -4 & -1 \\ 1 & 2 \\ 3 & 5 \end{bmatrix}$$

Matrix Equality

Two matrices are only considered equal if they have the same size or dimension or order and they have equal corresponding elements.

Cofactor of an Entry of a Matrix

A cofactor of an entry of a matrix is the same as the cofactor of the same entry in the determinant of the matrix and thus, is defined only for **square matrices**.

Sign convention: Excel Review Center

$$\begin{bmatrix} + & - \\ - & + \end{bmatrix} \begin{bmatrix} + & - & + \\ - & + & - \\ + & - & + \end{bmatrix}$$

Example:

Find the co-factor of 6 in the following matrix.

$$A = \begin{bmatrix} 1 & 2 & 0 \\ 2 & 1 & 4 \\ -4 & 2 & 6 \end{bmatrix}$$

The equivalent matrix is:

$$A = \begin{bmatrix} 1 & 2 \\ 2 & 1 \end{bmatrix}$$

The determinant of A is:

$$\det A = \begin{vmatrix} 1 & 2 \\ 2 & 1 \end{vmatrix} = (1)(1) - (2)(2) = -3$$

Thus, the cofactor of 6 is -3 .

Cofactor Matrices

A cofactor matrix is formed by replacing each element in the given matrix by its cofactor.

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Example:

Find cofactor matrix of A

$$A = \begin{bmatrix} 3 & 2 \\ 4 & 1 \end{bmatrix}$$

$$\text{Cofactor matrix of } A = \begin{bmatrix} 1 & 4 \\ 2 & 3 \end{bmatrix}$$

Inverse Matrices

Steps required to find for the Inverse Matrix (say A):

- Form the cofactor matrix of matrix A
- Form the transpose matrix of the cofactor matrix A
- Evaluate the determinant of matrix A
- Divide each element in the (matrix cofactor)^T by the determinant

Example:

Find the inverse matrix of A.

$$A = \begin{bmatrix} 1 & 2 & 0 \\ 2 & 1 & 4 \\ 4 & 2 & 6 \end{bmatrix}$$

$$\text{Cofactor matrix } A = \begin{bmatrix} -2 & 4 & 0 \\ -12 & 6 & 6 \\ 8 & -4 & -3 \end{bmatrix}$$

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(Cofactor matrix A)^T

$$= \begin{bmatrix} -2 & -12 & 8 \\ 4 & 6 & -4 \\ 0 & 6 & -3 \end{bmatrix}$$

$$\det A = \begin{vmatrix} 1 & 2 & 0 \\ 2 & 1 & 4 \\ 4 & 2 & 6 \end{vmatrix} = 6$$

$$\text{Thus, } A^{-1} = \frac{1}{6} \begin{bmatrix} -2 & -12 & 8 \\ 4 & 6 & -4 \\ 0 & 6 & -3 \end{bmatrix}$$

Determinants

Determinant is associated with square matrices (i.e. the number of rows = the number of column) only. Every determinant can be associated with a real number.

Properties of Determinants:

- If the rows of one determinant are the same as the columns of another, and in the same order, the two determinants are equal.

$$\det A = \begin{vmatrix} 1 & 2 & 0 \\ 2 & 1 & 4 \\ 4 & 2 & 6 \end{vmatrix}$$

$$\det A = \begin{vmatrix} 1 & 2 & 4 \\ 2 & 1 & 2 \\ 0 & 4 & 6 \end{vmatrix} = 6$$

- If two columns (or rows) of a determinant are interchanged, the value of the resulting determinant is equal to the negative of the value of the given determinant.

$$\det A = \begin{vmatrix} 1 & 2 & 0 \\ 2 & 1 & 4 \\ 4 & 2 & 6 \end{vmatrix}$$

$$\det A = - \begin{vmatrix} 2 & 1 & 4 \\ 1 & 2 & 0 \\ 4 & 2 & 6 \end{vmatrix} = 6$$

- If two columns (or rows) of a determinant are identical, the value of the determinant is zero.
- If the elements of a column (or row) of a determinant are multiplied by k , the value of the determinant is multiplied by k . **Continuation on next page....**